

Air University



Organization & Assembly (Lab)

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Lab Task # 10

Problem:

CASE AX

<0: put -1 in BX

=0: put 0 in BX

>0: put 1 in BX

END_CASE

If AX contains a negative number, put -1 In BX; if AX contains 0, put 0 In BX; if AX contains a positive number, put 1 In BX. Display it on the output screen.

Use JL, JE, JG, and JMP instructions to accomplish your task.

Code

```
.model small
.stack 100h

.code
main proc
    MOV AX, -5
    CMP AX, 0      ; Compare AX with 0
    JL  NEGATIVE   ; If AX < 0, jump to NEGATIVE
    JE  ZERO       ; If AX == 0, jump to ZERO
    JG  POSITIVE   ; If AX > 0, jump to POSITIVE

    NEGATIVE:
        MOV BX, -1 ; Set BX to -1
        JMP END_CASE ; Jump to end

    ZERO:
        MOV BX, 0   ; Set BX to 0
        JMP END_CASE ; Jump to end

    POSITIVE:
        MOV BX, 1   ; Set BX to 1
        JMP END_CASE ; Jump to end

    END_CASE:

    ; Prepare to print
    MOV AH, 02H     ; DOS function to display character
    CMP BX, 0
    JL  PRINT_NEG    ; If negative, print '-' first
    JE  PRINT_ZERO   ; If zero, print '0'

    ; Else positive, print '+' and then digit
    PRINT_POS:
        MOV DL, '+'  ; Print '+'
```

```
    INT    21H
    MOV    AX, BX        ; Move BX to AX for processing
    CALL   PRINT_NUM     ; Call subroutine to print number
    JMP    DONE

PRINT_NEG:
    MOV    DL, '-'       ; Print '-'
    INT    21H
    MOV    AX, BX
    NEG    AX            ; Make positive for printing
    CALL   PRINT_NUM
    JMP    DONE

PRINT_ZERO:
    MOV    DL, '0'       ; Print '0'
    INT    21H
    JMP    DONE

PRINT_NUM:                ; Subroutine to print AX as unsigned digit (assuming small number)
    ADD    AL, '0'       ; Convert to ASCII (for single digit, e.g., 1 -> '1')
    MOV    DL, AL
    MOV    AH, 02H
    INT    21H
    RET

DONE:
    ; Exit the program
    MOV    AH, 4CH       ; DOS function to terminate program
    INT    21H

main endp
end main
```

Explanation:

This assembly program demonstrates conditional branching to determine the sign of a number (AX) and then prints the corresponding sign (+, -, or 0) followed by the magnitude of the number in BX.

Code Explanation: Sign Check and Display

- Sign Classification (Conditional Jumps):** The program initializes AX to -5 and uses the **CMP** instruction followed by **JL** (Jump Less), **JE** (Jump Equal), and **JG** (Jump Greater) to classify the value of AX relative to zero. This acts as a multi-way IF/ELSE IF/ELSE construct.
- Mapping to Sign Code:** Based on the jump, the program sets the BX register to a corresponding sign code: -1 for negative numbers, 0 for zero, and 1 for positive numbers. Execution then jumps to **END_CASE**.
- Output Preparation:** The code prepares to display the result by comparing BX to 0 to determine which printing routine to execute: **PRINT_NEG**, **PRINT_ZERO**, or **PRINT_POS**.
- Printing Sign and Magnitude:**
 - If BX is **positive** (**PRINT_POS**), it prints '+', moves BX to AX, and calls **PRINT_NUM**.
 - If BX is **negative** (**PRINT_NEG**), it prints '-', then uses **NEG AX** to get the absolute value before calling **PRINT_NUM**.

- If BX is **zero** (PRINT_ZERO), it simply prints '0'.
5. **PRINT_NUM Subroutine:** This simple subroutine takes the unsigned single-digit magnitude in AL (via AX), converts it to its ASCII character by adding '0', and displays it using the DOS **Character Output function** (INT 21h, AH=02h).
6. **Result and Termination:** Since AX was initialized to -5, the code executes the NEGATIVE branch, sets BX to -1, jumps to PRINT_NEG, prints '-', negates AX to 1, calls PRINT_NUM to print '1' (the magnitude of -1), and finally exits using the DOS **Terminate Program function** (INT 21h, AH=4Ch).

Output

For Negative value

```
D:\>D:\test
-1
D:\>[ _
```

For Positive Value

```
D:\>D:\test
+1
D:\>[ _
```

For Zero

```
D:\>D:\test
0
D:\>[
```

THE END
